

Dan Brogan

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Project Portfolio Website: dan-brogan.github.io

EDUCATION:

University of Southern California, Los Angeles CA

M.S. Astronautical Engineering, Concentrations in Space Systems Design, Space Applications

Aug 2020 – Dec. 2021

GPA: 3.96

University of Rhode Island, Kingstown RI

B.S. Mechanical Engineering, Minors in Robotics Engineering, Mathematics

Aug 2016 – May 2020

GPA: 3.94

WORK EXPERIENCE:

Fractal Robotics, Bristol RI | *President, Lead Product Design Engineer*

June 2022 – Present

- Founded and led a tech startup developing a 5-axis 3D printer and slicer software to advance additive manufacturing
- Conducted dozens of customer interviews to identify needs that helped inform product design decisions
- Designed and verified hardware with hand calculations, modeling the entire assembly in CAD (Autodesk Inventor), use of GD&T practices, and iterative testing. Developed motion and temperature control firmware in Arduino (C++) for prototyping. Developed 5-axis slicer application (Python) for generating g-code
- Manufactured all custom components in-house using CNC machining, lathes, and 3D printing, while sourcing COTS parts from reliable vendors
- Open-sourced all hardware designs and slicer source code on fractalrobotics.com. Building and leading an international maker community around 5-axis 3D printing innovation through online forum
- Actively engaged in the Rhode Island startup community through founder networking and collaboration events

Roger Williams University, Bristol RI | *Adjunct Physics Professor*

Jan 2022 – Present

- Develops lectures and lab activities to engage students in college-level Physics and Astronomy courses

Bristol Municipal Career & Technical Academy, Bristol RI | *STEM Educator*

Jan 2023 – Present

- Designed and delivered hands-on 7-week workshops for middle and high school students, including a “Build Your Own 3D Printer” course and an “Engineering Career Exploration” program
- Led a 1-day CAD (Onshape) bootcamp introducing students to 3D design principles

USC Rocket Propulsion Lab, Los Angeles CA | *Analysis Engineer*

Jan 2021 – Dec 2021

- Collaborated with team members to design and verify aluminum retention rings for a space-rated solid rocket using hand calculations (MATLAB) and finite element analysis (ANSYS). Participated in critical design reviews and assisted with static fire test in the Mojave Desert
- Evaluated the design of carbon-composite rocket fins with stainless-steel leading edges under hypersonic launch conditions using hand calculations (MATLAB) and finite element analysis (ANSYS Composite PrepPost (ACP)). Contributed analysis results to team discussions and design iterations

USC Space Engineering Research Center, Los Angeles CA | *Structures Engineer*

Aug 2020 – Dec 2020

- Collaborated with research team members to perform analysis and redesign of flight vehicle structures for remote-controlled lunar lander test vehicle using hand calculations (MATLAB) and finite element analysis (ANSYS)

NASA RI Space Grant, Kingstown RI | *Principal Investigator*

Aug 2019 – May 2020

- Collaborated with Drs. Jouaneh and DiFilippo on research into computer vision for non-destructive electronic disassembly
- Created and annotated a dataset of cross-recessed screw fasteners to train a deep learning detection model
- Performed trade study of seven hyperparameter configurations to optimize detection accuracy and training efficiency
- Co-authored a peer-reviewed publication on deep learning methods for robotic disassembly and servicing applications

Lockheed Martin, Middletown RI | *Mechanical Engineering Intern*

Jan 2019 – Aug 2019

- Coordinated partnerships with small businesses on several SBIR projects
- Technical Lead and Small Business Coordinator for hydrostatic composite overwrapped pressure vessel (COPV) proposal. Used hand calculations and finite element analysis with ANSYS composite PrepPost (ACP) to verify the design against hydrostatic buckling failure modes
- Held secret security clearance and represented Lockheed Martin at an SBIR conference at Skunk Works in Palmdale, CA to evaluate potential partnerships with small businesses
- Operated machine shop milling machine and SLA resin 3D printers

URI Artificial Intelligence Laboratory, Kingstown RI | *Technical Assistant*

Oct 2018 – May 2019

- Constructed and tested robot kits, hosted Arduino workshops, and helped facilitate URI’s public AI lab

Raytheon, Portsmouth RI | *Mechanical Engineering Intern*

May 2018 – Aug 2018

- Generated technical drawings for electronics cabinets using PTC CREO Parametric and Product Data Management software
- Helped design and generate bill of materials (BOM) for sulfur-hexafluoride filling station
- Designed parts with O-ring grooves for internal R&D project
- Generated complex 3D models for “Enhanced Hour-Glass Transducer” with PTC CREO Parametric which contributed to patent # PH12021552475A1

PUBLICATION:

- Brogan, D. P., DiFilippo, N. M., & Jouaneh, M. K. (2021). Deep learning computer vision for robotic disassembly and servicing applications. *Array*, 12, 100094.

PROJECTS:

Lunar Rover Concept Architecture: Robotic Remote-Sensing Scout (R2-S2) Sept 2021 – July 2022

- Designed a versatile lunar EVA-assist rover concept with teleoperated and autonomous modes, configurable subsystems (solar panels, wheel width, insulation), and mission scenarios to enhance crew safety and productivity on Artemis missions
- Presented concept to academic peers and Dr. Buzz Aldrin, and later presented at the 2022 NASA Exploration Science Forum (NESF) conference

Lunar South Pole Base Design Sept 2021 – Dec 2021

- Led a team project in USC's *Human Spaceflight* course (taught by former NASA astronaut Dr. Garret Reisman) to design "Jamestown Base," a lunar outpost near the south pole, integrating site selection, life support, power, and EVA operations
- Produced a comprehensive systems-level report aligning the base concept with a cislunar transportation Concept of Operations (ConOps), demonstrating feasibility of long-duration human presence on the Moon

Titan (Moon of Saturn) Entry Descent & Landing (EDL) Mission Design Sept 2020 – Dec 2020

- Led a team in USC's *EDL for Planetary Surface Exploration* course (taught by Dr. Anita Sengupta) to design a robotic mission delivering a UAV to Kraken Mare, Titan's largest hydrocarbon sea
- Simulated hypersonic, supersonic, and subsonic EDL phases in MATLAB, producing a full mission architecture and technical report detailing vehicle sizing, entry conditions, and communication systems

3-Body Problem Orbital Mechanics Numerical Solver Nov 2020 – Dec 2020

- Extended a homework assignment from USC's *Orbital Mechanics* course (taught by Dr. Ryan Park) into a MATLAB simulation for the 3-body problem, numerically integrating gravitational perturbations to compute orbital motion
- Designed MATLAB script to generate an animation of the orbital dynamics of the 3 bodies based on user inputs (masses, initial conditions, time span), later publishing the code on GitHub

NASA Venus Rover Mechanical Sensors Apr 2020 – May 2020

- Collaborated with Mitchell Brogan to design purely mechanical obstacle avoidance sensors for NASA's *Exploring Hell: Avoiding Obstacles on a Clockwork Rover* HeroX challenge
- Generated technical report detailing the high-level design of a sensor suite capable of detecting slopes, rocks, and valleys. Discussed environmental, geometrical, functional, and mass requirements

Turbo Jet Engine May 2019 – Aug 2019

- Fabricated and tested a turbo jet engine comprised of a turbocharger, combustion chamber, active oil lubrication and cooling system, fuel inlet, and ignition system

Pulse Jet Engine May 2018 – Aug 2018

- Fabricated and tested a pulse jet engine comprised of a combustion chamber, intake/exhaust pipes, and fuel inlet. Self-taught TIG welding to join stainless steel sheet metal components

SKILLS:

- **Engineering & Analysis:** Hand-Calculations, MATLAB, Finite Element Analysis (FEA), ANSYS Workbench, ANSYS Composite PrepPost (ACP)
- **CAD:** Autodesk Inventor, SolidWorks, Onshape, CREO Parametric
- **Software & Firmware:** Python, Arduino (C++), Robotics, Motion Control, PID Temperature Control Systems, Multi-Axis Calibration Routines, Simultaneous Control of Circuit Elements
- **Manufacturing & Prototyping:** CNC Machining, Lathe Machining, TIG Welding, Soldering, Vacuum Forming, Composites, Design for Manufacturing & Assembly (DFMA), Geometric Dimensioning & Tolerancing (GD&T)
- **Professional Skills:** Leadership, Teamwork, Teaching, Technical Communication, Microsoft Excel, Earned Value Management System (EVMS)

ASSOCIATIONS:

URI Astroneering, Kingstown RI | Founder, President Aug 2019 – May 2020

- Founded and led group for aspiring rocket scientists and like-minded people who share an interest in solar system exploration. Hosted workshops for making solid rockets from scratch, lectures in aerodynamics and rocket design, and experiments with closed-loop terrariums for interplanetary botany

URI American Society of Mechanical Engineers (ASME), Kingstown RI | President Jan 2017 – May 2019

- Revived previously defunct URI ASME student chapter. Organized design competitions, taught lectures and held discussions in a variety of Engineering and Technology topics. Organized field trips to Engineering companies

URI Engineering Council, Kingstown RI | Co-Chair May 2018 – May 2019

- Coordinated networking and inclusion events between URI student organizations including IEEE, SHPE, SWE, NSBE, ESW, AICHE, ASCE, and EMBS alongside Co-Chair Holly Robinson. Helped organize Engineering Week, delegating event planning among URI Engineering student organizations. Organized and hosted formal "Evening with Industry" speed-networking event between local Engineering employers and STEM students